TEXORA · SESSIONAL NOTES · PRACTICAL STUDY SHEET

Practical Study of Different Mechanisms: Feeding, Drawing-Off, Winding and Receiving, Driving and Stopping Mechanisms

The five supporting mechanisms that decide whether a knitting machine makes good fabric or waste: how yarn is fed (negative, positive and storage feeding), how fabric is drawn off and batched, how the machine is driven, and how its stop motions protect needles, yarn and fabric — with the storage feeder drawn in full.

TE-3102 · KNITTING TECHNOLOGY SESSIONAL · STUDY 3 · LEVEL 3 TERM 1

1. Object and method

Object: to identify, trace and explain the five mechanism groups of the laboratory machines — feeding, drawing-off (take-down), winding and receiving (batching), driving, and stopping — and to record the setting of each. These mechanisms decide fabric uniformity more than the knitting elements themselves: an uneven feed or take-down will spoil fabric from a perfect set of needles.

Method. For each mechanism: trace its path (yarn, fabric or power), name the parts in order, find its adjustment, and state in one line what goes wrong in the fabric when that adjustment is wrong. Section 7 collects the report record.

2. Yarn feeding mechanisms

2.1 Negative feeding — the needles help themselves

In the simplest arrangement the yarn runs from cone through guides and a tension device straight to the carrier, and the needles **draw off whatever length the stitch cam demands**. The feed is negative because nothing is supplied positively: the length taken depends on yarn tension, package state and path friction, all of which vary — cone to cone, outside to inside of a cone, and with every knot or slub. The result is stitch-length variation, showing as barré and uneven course density. Negative feeding survives on simple machines and the hand flat machine, where speeds are low.

2.2 Positive feeding — the machine meters the yarn

A positive feed unit delivers a **measured length of yarn per machine revolution**, regardless of the tension at which the needles take it. The common forms are:

- **Tape-and-wheel feeders** — yarn nipped between a rotating feed wheel and a moving tape; the wheel's surface speed, set by change pulleys or a variable drive, meters the yarn. All feeders are driven from one shaft, so every feeder delivers in proportion.
- **The storage feeder** — the modern standard, and the unit fitted in this laboratory (Figure 1). A rotating arm winds yarn from the cone onto a smooth drum as a reserve of separated coils; the needles withdraw yarn **over the end of the drum**, coil by coil, at almost constant low tension. Package irregularities never reach the needles — the drum absorbs them, refilling as fast as it empties. Delivered length per revolution is set by the winding shaft's drive ratio, common to all feeders.

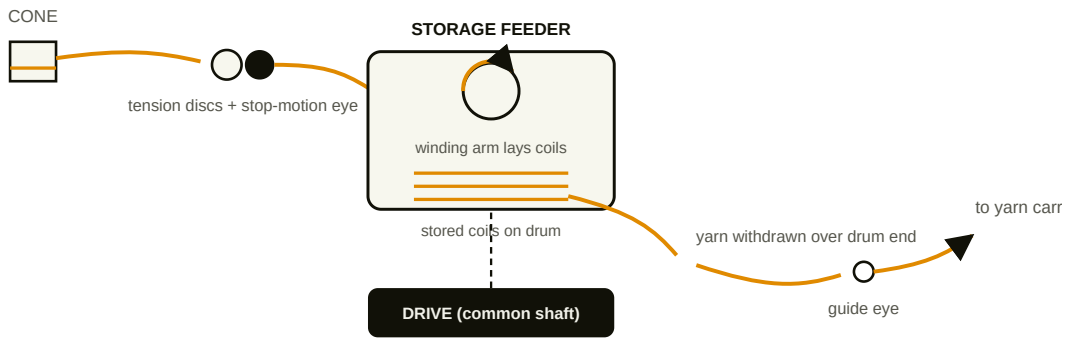


Figure 1 — The storage (accumulator) yarn feeder. Yarn passes the tension discs and stop-motion eye, is wound by the rotating arm into stored coils on the drum, and is withdrawn over the drum end at near-constant tension to the carrier. Metering is positive (drive-set); tension at the needles is decoupled from the package.

2.3 What to observe and record on feeding

- **Which system is fitted** — negative, tape/wheel positive, or storage; number of feed units and whether all work.
- **The delivery setting** — pulley step or drive ratio in use; verify by measurement: run a counted number of revolutions and measure the length delivered per feeder. Delivered length ÷ needles knitting at that feeder should equal the stitch length (compare Study 5).
- **Input tension** — tension-disc spring setting; on a storage feeder this only has to wind the drum reliably — it does *not* set knitting tension (a favourite viva point).
- **Warp machine** — feeding is **let-off from the beams**: record negative (beam braked) or positive (beam driven) let-off, and the run-in per bar (Study 1). **Flat machine** — feeding is by the travelling carrier, effectively negative; record the mast tensioner setting and carrier in use.

3. Drawing-off (take-down) mechanism

New loops must be pulled from the needle line at exactly the rate courses are produced, under tension high enough to keep loops open and fabric controlled, but never enough to stretch loops or rob yarn from the forming course (**robbing back**). On the circular machine the tube passes down inside the cylinder to **take-down rollers** (a driven steel roller against rubber-covered rollers), driven from the main drive through change gears. On the flat machine, rollers or a weighted comb do the same work; on the warp machine, geared **take-up rollers** draw fabric from the trick plate.

- **Identify:** rollers and covering, nip adjustment (spring or weight), drive and change gears, and any spreader keeping the tube open before the nip.
- **Record:** the setting in use and how it is changed. Course spacing is set jointly by course rate and take-down rate, so **courses per cm is a take-down result as much as a knitting result** (Study 5 measures it).
- **Faults:** too little take-down — fabric bunches at the needles, loops distort and press off; too much — loops stretched vertically, courses per cm fall, width narrows, yarn robbed from the forming course. Slipping rollers (worn covering, weak nip) give irregular density down the piece.

4. Winding and receiving (batching) mechanism

Below the take-down rollers the circular machine winds fabric onto a **batching roll**, surface-driven in a cradle or on driven rollers, so surface speed and winding tension stay roughly constant as the roll builds while its rotational speed falls. Some machines plait (fold) the fabric into a box instead, avoiding winding stretch and roll marks. Flat-machine panels are received by hand into a bin; warp fabric passes over guide rollers to a large front batching roll, often past an inspection light.

- **Identify:** batching roll and drive (surface or centre), cradle/arms, doffing arrangement, and — where fitted — the metre counter and inspection point.
- **Record:** roll or plait receiving; target roll size/weight per doff; how a doff is made.
- **Faults:** telescoping or hard/soft roll edges (uneven winding tension), creases wound in (fabric not spread before the nip), oil or dirt picked up at receiving — faults made here cannot be removed later.

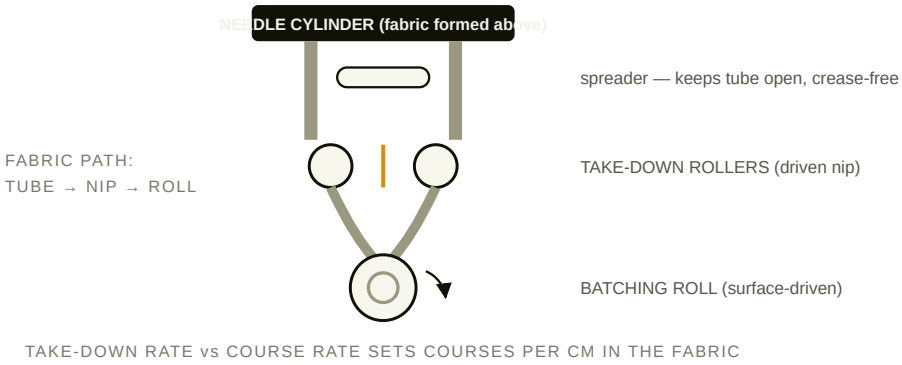


Figure 2 — Drawing-off and receiving on the circular machine. The spreader opens the tube, the driven roller nip draws fabric down at the set rate, and the surface-driven batching roll winds it on at constant surface speed.

5. Driving mechanism

Trace the power path: **motor** (plate rating and speed) → **belt, chain or direct coupling** → **main shaft** → the branch drives: cylinder (or cam-ring) drive, feeder-drive shaft (Section 2 — why feeder delivery stays proportional to machine speed), take-down gearing (Section 3), and on the warp machine the shaft timing needle bar, guide bars and sinkers together. Modern machines add a **variable-speed drive** and an **inching (jog) control**; older machines use stepped pulleys. On the flat machine the drive is the carriage traverse drive (or the knitter's hand).

- **Record:** motor plate data; speed range and speed in use; inching control location; direction of rotation (frame arrow).
- **Key viva idea:** feeders and take-down are driven from the same main drive as the needles, so their rates rise and fall with machine speed — stitch length and course density stay constant through speed changes.

6. Stopping mechanisms

A knitting machine at full speed forms thousands of loops a minute; a fault left running for ten seconds can destroy a needle set or a roll of fabric. Stop motions watch the machine and stop it — and signal where — the moment a fault begins. The laboratory machines carry most of the following; find and identify each, and note how it is tested (Study 4).

STOP MOTION	WHAT IT WATCHES	HOW IT ACTS
Yarn-break / yarn-end detector	Each yarn on its way to the feeder: a drop wire, feeler finger or optical sensor held up by the running yarn	Yarn gone → feeler falls (or beam clears) → switch cuts the drive and lights the signal lamp for that feeder; on some machines the corresponding yarn cutter also fires

Knot / slub detector	Yarn thickness at a slotted gate before the feeder	An oversize knot cannot pass the gate; the gate's movement trips the stop and the yarn is cut, so the knot never reaches the needles
Needle-break detector	The fabric just below the needle line, or the needle circle itself (feeler resting on the fabric/needle heads)	A broken needle leaves a wale unknitted (a ladder) or a hook fragment in the fabric; the feeler drops into the gap or is struck and stops the machine before the fault runs on
Press-off / fabric detector	Presence of fabric under proper take-down tension below the cylinder	If the fabric presses off (falls from the needles) or take-down fails, the detector stops the machine instead of letting loose fabric wrap the rollers or needles re-knit air into a needle-breaking tangle
End-of-roll stop; emergency stop	Batching roll size/metre counter; the operator (red mushroom buttons, interlocked guards)	Stops or signals at the set length so rolls are uniform; emergency stop cuts the drive and brakes at once, as does opening an interlocked guard

Fail-safe principle: the *fault itself* completes the action — yarn present holds the detector off, yarn absent lets it stop the machine. A broken detector wire therefore stops the machine rather than disabling the protection, and a machine that "will not start" is often a stop motion correctly reporting a real fault (Study 4).

7. What the lab report must contain

SECTION	CONTENT
Feeding	System fitted, parts in order along the yarn path, delivery setting, and your measured delivery check (length per revolution)
Drawing-off	Roller arrangement, loading and drive; setting in use; one fault of too much and one of too little take-down
Winding/receiving	Roll or plait, drive type, doffing method and roll target
Driving	The full power path traced, motor plate data, speed in use, inching control location

Stopping	Every stop motion found, what it watches, and how each was tested or demonstrated

8. Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. Why is negative feeding unable to hold stitch length constant even on a perfectly adjusted machine?
2. On a storage feeder, the input tension is doubled. What changes at the needles — and what does not? Explain from the mechanism.
3. How would you verify a positive feeder’s delivery setting by measurement, and what should the result equal?
4. Courses per cm in a fabric fall after the take-down setting is increased, although the machine speed and feeder setting are untouched. Explain.
5. State the fail-safe principle of stop motions and one practical consequence of it for fault finding.

Answer key

1. Nothing meters the yarn: the needles draw against a resistance that varies with package size and position, guide friction, knots and yarn irregularity. Delivery follows the varying resistance, not a set length, so stitch length — and fabric weight and density — wanders.
2. Almost nothing at the needles. Higher input tension only winds the drum more firmly; withdrawal over the drum end is still coil-by-coil from the free reserve, so knitting tension and delivered length (set by the drive ratio) are unchanged. Only if tension were so low that the drum emptied would feeding collapse to negative until the reserve rebuilt.
3. Run a counted number of revolutions and measure one feeder’s delivered length. Divided by the needles knitting at that feeder it should equal the **stitch length** — cross-check against unravelling (Study 5).
4. Course density is course production relative to draw-off. Faster take-down pulls each course further from the needle line before the next forms, spacing the same courses over a longer length — courses per cm fall, loops draw taller, fabric narrows. Take-down is a density control, not merely a removal device.
5. The fault condition itself produces the stop: running yarn holds the detector *off*; absence of yarn — or a broken detector circuit — lets it act. Consequence: a machine that will not start should first be checked for a genuine reported fault (yarn broken, fabric pressed off) before the electrics are blamed.